Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
#include<stdio.h>
void func1(int *arr){
  for(int i=1; i<=4;i++)
    (i-1)[arr] = (i%4);
}
void func2 (int a[1] , int arr){
  a[0] = arr + 2;
}
int main(){
  int arr[5] = {1,1,1,1,1};
  func1(arr+1);
  for(int i=0; i<5; i++)
        printf("%d ", i[arr] );
  printf("\n");
  func2(arr, *arr);
  for(int i=0; i<5; i++)
        printf("%d ", i[arr] );
  return 0;
}
```

Output
1 2 3 4
1 1 1 1

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
  int temp = arr + 2;
  a = &temp;
}
```

Explain your answer

Yes since arr+2 stores the value of address. But a here is giving the value of address of the address ∴ So answer changes.

Q. 2: Write the output of the following sequence of instructions **long arr2[60]; int arr1[50], a1=(&arr2[40]), a2=(&arr2[30]), b1=(&arr1[40]), b2=(&arr1[30]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

$a_1 = arr2 + 40$
$a_2 = arr2 + 30$
$b_1 = arr1 + 40$
$b_2 = arr1 + 30$

$a_1 - a_2 = \cancel{arr_2} + 40 - \cancel{arr_1} - 30 = 10$

$b_1 - b_2 = \cancel{arr1} + 40 - \cancel{arr1} - 30 = 10$

Out put :-
10 10